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Clustering algorithm for experimental datasets using global sensitivity-based affinity propagation (GSAP)

Yiru Wang[a,b], Chenyue Tao[a,b], Zijun Zhou[a,b], Keli Lin[a,b], Chung K. Law[a,c], Bin Yang[a,b,*]

[a] Center for Combustion Energy and Department of Energy and Power Engineering, Tsinghua University, Beijing 100084, PR China
[b] Key Laboratory for Thermal Science and Power Engineering of MOE, International Joint Laboratory on Low Carbon Clean Energy Innovation, Tsinghua University, Beijing 100084, PR China
[c] Department of Mechanical and Aerospace Engineering, Princeton University, Princeton, NJ 08544, USA

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ABSTRACT

To minimize the uncertainty of the parameters in combustion kinetics models, Bayesian methods are commonly used for uncertainty constraints based on experimental data. With the rapid and substantial growth of experimental data, using all the experimental data for optimization is not only redundant and time-consuming, but it could also lead to data consistency problems. In this work, the global sensitivity-based affinity propagation method (GSAP) is proposed to cluster experimental datasets and to select representative experimental conditions. Specifically, the global sensitivity coefficient is first obtained through an analysis to characterize the sources of uncertainty in the kinetic model under different experimental conditions. The similarity coefficient, which is defined based on the global sensitivity, measures the resemblance between two experimental conditions. By exchanging messages calculated from similarity, affinity propagation enables the experimental dataset to be automatically clustered into several classes without specifying the number of classes in advance. This method innovatively introduces the consideration of model and experimental uncertainty under different conditions to obtain better optimization results. The correctness and effectiveness of the method are validated through clustering and optimizing on a laminar flame speed dataset of common C_0–C_4 fuels. The dataset consisting of 288 experimental conditions has been automatically clustered into 27 categories, and an exemplar of each category is given. These exemplary conditions reflect the dominant chemistry behind their cluster. At the same time, these conditions have larger model prediction uncertainty and smaller experimental uncertainty to provide better Bayesian constraints. The uncertainty of the model parameters after Bayesian optimization is effectively constrained. The average uncertainty of model predictions across the dataset is reduced from 30 % to 10 % using only 27 exemplar conditions for optimization. While selecting experimental data for model optimization, the clustering strategies provided by this method also, in turn, help understand its underlying chemical essence.

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1. Introduction

Combustion kinetics models are important tools in the design and optimization of combustion systems. However, current combustion kinetics models still have large uncertainties due to the uncertainties in model parameters and the absence of reaction pathways. With the former, inverse uncertainty quantification [1] is usually adopted to constrain the uncertainty space and reduce the model prediction uncertainty using experimental data. Frenklach and co-workers proposed a series of methods for model optimization and data collaboration [2–11]. Turányi et al. introduced methods to quantify the Arrhenius parameters' uncertainty and conducted optimization on several combustion kinetics models [12–16]. Sheen and Wang [17–21] developed the Method of Uncertainty Minimization using Polynomial Chaos Expansions (MUM-PCE), which could significantly reduce the computational cost spent on the model evaluation. Many methods based on Bayesian inference have also been used to infer the posterior distribution of the kinetics parameters [22]. Wang et al. proposed the Artificial Neural Network-Markov Chain Monte Carlo (ANN-MCMC) method [23], which significantly reduces the computational cost of Monte Carlo sampling.

However, with the continuously increasing number of experimental data, using all the data for optimization is tedious and time-consuming, and may lead to consistency problems. In order to address these challenges, methods for data classification and

* Corresponding author.
E-mail address: byang@tsinghua.edu.cn (B. Yang).

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selection of representative experimental conditions are needed, which aim to identify the most critical experiment conditions that contribute the most to the overall model uncertainty and prioritize them in the optimization process. There are also many explorations concerning data consistency and information content. Pioneering work on the consistency of experimental data and minimizing the number of experimental data points can be traced back to the solution mapping method and data collaboration approach proposed by Frenklach et al. [2,8,24–26]. And by using collaborative data processing, Bound-to-Bound Data Collaboration (B2B-DC) is proposed by Frenklach et al. to handle the consistency of dataset. A rigorous measure of dataset consistency is designed based on constrained optimization, utilizing solution mapping tools and robust control algorithms [8]. This method can quickly screen the dataset and deliver a feasible dataset for model optimization. Furthermore, consistent data may also provide different information, hence how to measure the information contained in the data is also crucial to the selection of data points. Li et al. [27] proposed the concept of sensitivity entropy to measure the degree of dispersion of uncertainty sources. Sheen and Manion [28] and Valko et al. [29] developed differential entropy for experimental design.

However, the sensitivity entropy-based studies focus more on the information content of a single experimental condition. A well-developed clustering and experimental condition selection algorithm needs to consider the linkages between experimental conditions. Consequently, it is necessary to have a similarity measure and clustering algorithm. These tools are essential for identifying patterns and relationships in the experimental data, and grouping them into clusters based on their similarity. By doing so, they enable the selection of the most representative data points for the optimization process, reducing the computational cost while maintaining accuracy. Moreover, similarity measures and clustering algorithms can also help identify outliers and noisy data points, which can significantly affect the accuracy and reliability of the resulting model. The "correlation" among experiments is considered by introducing "bias" by Frenklach et al. [30] and further used in identifying the relationships of the experimental data for uncertainty quantification. Wang et al. developed the surrogate model similarity (SMS) method [31], which calculates a similarity coefficient between different models' predictions based on their characteristic coefficients. This coefficient represents how close two models' predictions are and hence can help find experimental targets that are more effective in reducing uncertainty and avoid selecting data with redundant information. A similar similarity measure was proposed by Lin et al. [32]. This method uses active subspace parameters as feature to calculate the similarity between conditions. However, these two methods only provide a metric between two conditions and are suitable for hand-picking experimental conditions with small amounts of data. For a large-scale dataset, more suitable clustering algorithms are still needed to perform clustering to identify the reaction characteristics under different conditions. Clustering results not only reduce the computational effort of Bayesian optimization, but also provide a benchmark dataset for model validation. Kovacs et al. [33] proposed a screening method for active parameters and experimental subsets that combines principal component analysis, sensitivity coefficients and model and experimental uncertainties, achieving a computational acceleration of 4–7 times. Valko et al. [34] introduced the Weighted Pair Group Method with Arithmetic Mean clustering algorithm for the identification of homogeneous chemical kinetic regimes of methane-air ignition. However, one of the above two methods cannot be used for clustering and the other cannot provide cluster centers.

In this study, a clustering and exemplars selection algorithm called global sensitivity-based affinity propagation (GSAP) is proposed. The algorithm allows to automatically divide the experimental dataset into several clusters based on the principle of net similarity minimization without specifying the number of clusters in advance. Global sensitivity analysis is employed in this algorithm to measure the similarity between experimental conditions. The affinity propagation algorithm automates the clustering and exemplar selection based on similarity to serve model optimization. It can significantly constrain the uncertainty of the entire dataset by using only a small number of exemplars generated by the algorithm for optimization.

2. Methodology

The overall logic of this algorithm (GSAP) is shown in Fig. 1 and as follows: (1) Under various experimental conditions (data points), use the global sensitivity analysis to obtain the global sensitivity of each condition as the feature. ANN–HDMR method [35–38] is adopted in this work. (2) Use the global sensitivity as input, affinity propagation enables spontaneously clustering and exemplar selection of the dataset.

In this work, we focus on the pre-exponential factor, which is assumed to be lognormally distributed:

$$lnx_i \sim N\left(lnx_i^0, \left(\frac{lnf_i}{2}\right)^2\right) \tag{1}$$

where x_i and f_i denote the ith input parameter and the corresponding uncertainty factor, respectively, and f_i is defined as the ratio of the maximum of ith input parameter x_i^{max} and the nominal value x_i^0. The input parameters are then normalized as follows:

$$x_i^* = \frac{\ln x_i - \ln x_i^0}{\ln f_i} \tag{2}$$

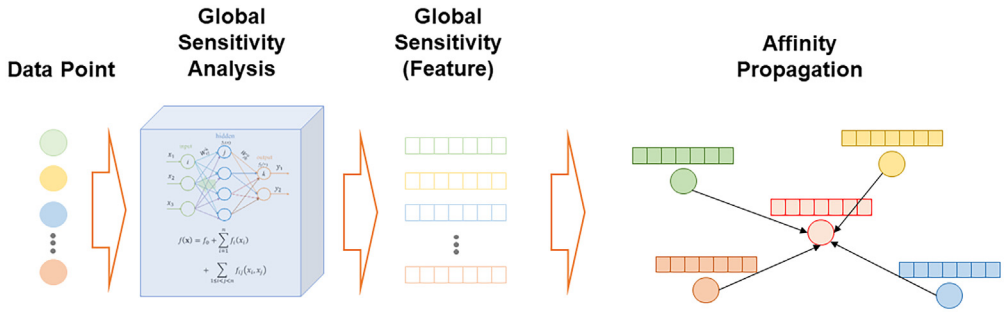


Fig. 1. Framework diagram of the affinity propagation algorithm based on global sensitivity analysis (GSAP).

2.1. Global sensitivity analysis

First, the local sensitivity analysis is used to screen the sensitive reactions under the corresponding conditions. Reactions with a local sensitivity greater than 10^{-3} for each experimental condition are screened out. Only filtered active reactions will have global sensitivity parameters calculated to reduce computational complexity and the global sensitivities of inactive reactions are set to be 0. Then the global sensitivity coefficient of the sensitive reactions in the mechanism is calculated using a second-order high dimensional model representation:

$$F(x) \approx F_0 + \sum_{1\le i\le n} F_i(x_i) + \sum_{1\le i<j\le n} F_{ij}(x_i, x_j) \tag{3}$$

where F_0 is the expectation of the model prediction, x_i and x_j the ith and jth inputs, and F_i and F_{ij} the first-order and second-order component functions of the HDMR model. The component functions are constructed from orthogonal polynomials proposed by Li et al. [37,39,40]:

$$F_i(x) \approx \sum_{r=1}^{k} \alpha_r^i \phi_r^i(x_i) \tag{4}$$

$$F_{ij}(x_i, x_j) \approx \sum_{p=1}^{l}\sum_{q=1}^{l'} \beta_{pq}^{ij} \phi_p^i(x_i)\phi_q^j(x_j) \tag{5}$$

where ϕ_r^i is the orthogonal polynomial, α_r^i and β_{pq}^{ij} the characteristic coefficients, which can be calculated with the Monte Carlo method:

$$\alpha_r^i \approx \frac{1}{N}\sum_{s=1}^{N} F(\boldsymbol{x}^s)\phi(x_i^s) \tag{6}$$

$$\beta_{pq}^{ij} \approx \frac{1}{N}\sum_{s=1}^{N} F(\boldsymbol{x}^s)\phi_p(x_i^s)\phi_q(x_j^s) \tag{7}$$

where $\boldsymbol{x}^s$ denotes the sth input sample; x_i^s and x_j^s are the ith and jth components of $\boldsymbol{x}^s$; and N is the number of input samples generated by Monte Carlo random sampling. The variance-based global sensitivities are then derived from the characteristic coefficients:

$$S_i = D_i/D = \left(\sum_{r=1}^{k}\left(\alpha_r^i\right)^2\right)/D \tag{8}$$

$$S_{ij} = D_{ij}/D = \left(\sum_{p=1}^{l}\sum_{q=1}^{l'}\left(\beta_{pq}^{ij}\right)^2\right)/D \tag{9}$$

where S_i and S_j denote the first- and second-order global sensitivities of the corresponding reactions, respectively, and D is the variance of the model predictions. Since the Monte Carlo method needs to generate extensive amount of samples, artificial neural networks (ANN) are used to speed up the simulation [38,41,42].

2.2. Global sensitivity-based affinity propagation

Affinity propagation [43] is used to identify exemplars among data points and form clusters of data around these exemplars. All data points are considered potential exemplars in the initial state. By exchanging messages between the data points, a set of exemplars and clusters emerges eventually. A dataset is then described using a small number of exemplars, which are identified as those most representative of other samples. This provides the possibility to cluster complex experimental datasets and select representative experimental conditions for Bayesian optimization to reduce the uncertainty of the model on the entire dataset.

The core of the algorithm is message exchange between pairs of data points. Two categories of messages are sent between the data points: the first is the responsibility $r(i,k)$, which is sent from cluster member i to candidate exemplar k, indicating how well-suited the data point i should be a member of the cluster candidate k; the second is the availability $a(i,k)$, which is sent from candidate exemplar k to potential cluster member i, indicating how appropriate candidate k would be as an exemplar. Formally, the responsibility and availability are given by:

$$r(i,k) \leftarrow s(i,k) - \max\left[a(i,k') + s(i,k'), \forall k' \neq k\right] \tag{10}$$

$$a(i,k) \leftarrow \min\left[0, r(k,k) + \sum_{i' \text{ s.t. } i'\notin\{i,k\}} r(i',k)\right] \tag{11}$$

where $s(i,k)$ is the similarity between data points i and k. To begin with, all the r and a are set to zero. The responsibilities and availability are recomputed and updated in each iteration based on the values from the previous iteration until convergence. Based on (10) and (11), the similarity between any two data points needs to be given when calculating the two categories of information passed in each iteration. In this algorithm, similarity is defined by the negative Euclidean distance of the global sensitivity of two data points:

$$s(i,k) = -||GS_i - GS_k|| \tag{12}$$

where $s(i,k)$ is the similarity between data point i and data point k, and $\boldsymbol{GS_i}$ and $\boldsymbol{GS_k}$ indicate the global sensitivity coefficients vector of the data point i and k, respectively. The similarity of a data point to itself is called a "preference", which is not unity, and is an input parameter that can be user-defined. Data points with greater "preference" are more inclined to be an exemplar. It will be discussed in more detail in Section 3.3. Negative Euclidean distance with global sensitivity is used in this study to define the similarity between the data points.

Dataset clustering and selection of representative experimental conditions could be considered as an exemplar-based clustering problem. The goal is to find the clustering and exemplar selection strategy that maximizes the net similarity, and can be abstracted as a mathematical problem:

$$\text{NetSimilarity}(c) = \sum_{ik} c_{ik} s(i,k) \tag{13}$$

where c_{ik} indicates whether or not point k is the exemplar of point i and $c_{kk} = 1$ indicates that point k is an exemplar.

2.3. Optimization method

The convergence of the affinity propagation algorithm yields not only a clustering strategy but also an exemplar for each cluster, which holds potential for model optimization. To evaluate the impact of these exemplars on model optimization, it is essential to optimize the model and compare the distribution of uncertainty before and after the optimization process. In this work, the ANN-MCMC proposed by Wang et al. [23] is used to optimize the model parameters.

Markov Chain Monte Carlo (MCMC) is a powerful Bayesian inference method widely used for exploring complex and high-dimensional probability distributions. The main goal of MCMC is to generate a set of samples from a target distribution that is difficult to sample directly, which is the posterior distribution of model parameters in this work:

$$p_{\text{post}}(\boldsymbol{x}|d) = \frac{p_{\text{prior}}(\boldsymbol{x})\pi(\boldsymbol{x};d)}{\int p_{\text{prior}}(\boldsymbol{x})\pi(\boldsymbol{x};d)d\boldsymbol{x}} \propto p_{\text{prior}}(\boldsymbol{x})\pi(\boldsymbol{x};d) \tag{14}$$

where $p_{post}(\boldsymbol{x}|d)$ denotes the posterior distribution of model parameters $\boldsymbol{x}$ under the given data d. $p_{prior}(\boldsymbol{x})$ is the unoptimized prior distribution, which is assumed to be a normal distribution introduced in Eq. (2). $\pi(\boldsymbol{x};d)$ is the likelihood function describing the probability of observing the data given the model parameters:

$$\pi(x;D) = \frac{1}{\prod_{i'}\sqrt{2\pi\sigma_{i'}^2}} \exp\left[-\frac{1}{2}\sum_{i'}\left(\frac{lnF_{i'}(x) - lnd_{i'}}{\sigma_{i'}}\right)^2\right] \quad (15)$$

where $F_i(x)$ is the simulation result and σ_i is the corresponding uncertainty.

By constructing a Markov chain that has the target distribution as its equilibrium distribution, MCMC algorithms can draw samples from this distribution by iteratively proposing and accepting or rejecting new states of the Markov chain [44]. The samples obtained from MCMC are used for estimating posterior distributions. To save sampling time, ANN surrogate model is utilized in place of combustion kinetics simulations [23].

2.4. Case study

The USC Mech II, which is a detailed kinetic mechanism describing high-temperature chemistry of H_2/CO and C_1–C_4 hydrocarbons with 111 species and 784 reactions, is employed in this work. The model has been extensively validated and is applicable to a wide variety of combustion scenarios including ignition delay time, laminar flame speed, species profiles in flow reactor, shock tube and burner stabilized flame.

The uncertainty factors for the Arrhenius pre-factors in USC Mech II are provided in the supplementary material of Ref [19]. The reason for choosing this mechanism is that it has a relatively complete reaction path of the core mechanism and has not been optimized for a specific data set. This research focuses on the clustering of laminar flame speed data and using these data for model optimization. Therefore, H_2, CH_4, CH_3OH, C_2H_2, C_2H_4, C_2H_6, pC_3H_4, C_3H_6, C_3H_8, C_4H_6, 1-C_4H_8, C_4H_{10}, as contained in USC Mech II, were selected as fuels. The equivalence ratio varied between 0.6–2.0, which is also a common range in the USC Mech II validation set. The pressure was set at 1, 2, and 5 atmospheres, while the cold side temperature was set to 300 K. Based on the above combinations of fuel, pressure, and equivalence ratios, a total of 288 experimental conditions were generated. Details can be found in ***Supplementary Material 1***. The kinetic simulation is conducted on the Cantera platform [45] using the free flame reactor. The width is set to be 0.01, and the relative and absolute tolerance are 10^{-7} and 10^{-11}, respectively. The grid is set to be automatically refined with criteria parameters: ratio=10, slope=0.05, curve=0.1, prune=0.04.

3. Results and discussion

3.1. Clustering results and visualization

We next utilize global sensitivities as input to automatically cluster the 288 conditions into 27 distinct categories through the application of affinity propagation. An exemplar is assigned to each category, effectively representing the corresponding cluster.

The distribution of global sensitivity under the exemplar conditions for each category is displayed in Fig. 2. To enhance clarity, only the significant reactions with high global sensitivities are presented. Throughout the remainder of the text, the reaction index is shortened to R. The initial 6 reactions (R0 to R17) pertain to the H_2 sub-mechanism. Among them, R0: H + O_2 <=> O + OH, is one of the most important elementary reactions, and as such is significant in almost all of the categories. The following 5 reactions, R33 to R43, belong to sub-mechanism of CO, and is also an important part of the core mechanism. The global sensitivity of the related reactions is naturally not as large as that of the H_2 sub-mechanism reactions, but it is still widely distributed in different cases. R53 to R119 are related to reactions of some C_1 free radicals, such as methyl, methylene, etc. Combustion of hydrocarbons, especially the high temperature reaction pathway, is a process in which the carbon chains are broken and small free radicals are generated. Consequently, the reactions between these C_1 radicals and H, O, OH, HO_2 etc. radicals are very important and ubiquitous in different fuels and experimental conditions. It is seen that except for the exemplar of the clusters 0 to 2, these chemical reactions have significant global sensitivity in other cases. This is because clusters 0 to 2 represent the reaction system of hydrogen and methanol. As hydrogen is a zero-carbon fuel, its reaction system is naturally independent of the reaction path of C_1 radicals. The reaction path of methanol is also different from that of common alkanes and alkenes. Among them, R108: CH_3 + HO_2 <=> CH_3O + OH is the

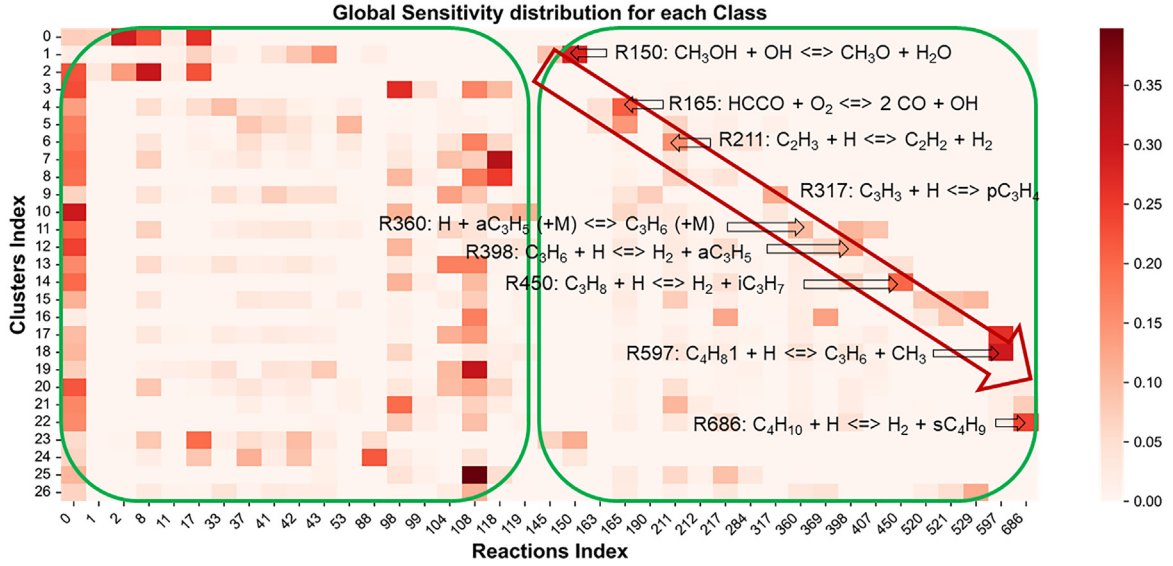


Fig. 2. Global sensitivity distribution of each exemplar. The left box shows the C_0–C_1 reaction shared by different conditions; the right box shows the fuel-specific reactions. The horizontal axis represents the index of reactions in the mechanism. Detailed reactions and their indexes are provided in ***Supplementary Material 2***.

Table 1
Selected exemplars based on Affinity Propagation.

Cluster index	Fuel	Pressure (atm)	Phi
0	H_2	1	1.0
1	CH_3OH	1	0.6
2	H_2	2	1.6
3	CH_4	2	1.6
4	C_2H_2	2	0.8
5	C_2H_2	2	1.6
6	C_2H_4	2	1.2
7	C_2H_6	2	1.0
8	C_2H_6	2	1.8
9	pC_3H_4	2	0.8
10	pC_3H_4	2	1.8
11	C_3H_6	2	1.0
12	C_3H_6	2	1.8
13	C_3H_8	2	0.8
14	C_3H_8	2	1.8
15	C_4H_6	2	1.2
16	C_4H_6	2	2.0
17	$1\text{-}C_4H_8$	1	1.8
18	$1\text{-}C_4H_8$	2	0.8
19	C_4H_{10}	2	0.6
20	C_4H_{10}	2	1.0
21	C_4H_{10}	2	1.4
22	C_4H_{10}	2	1.8
23	CH_3OH	2	1.0
24	CH_3OH	2	1.8
25	C_2H_4	5	1.8
26	C_4H_6	5	0.8

most critical one. This reaction has the highest average global sensitivity except for the H_2/CO sub-mechanism. This is very reasonable because this reaction promotes the activity of the whole reaction system. It was also shown in previous reaction kinetic studies carried out by Zhang et al. [46,47] that this reaction plays a crucial role in the prediction of ignition delay time, flame speed, species mole fraction, and as such is the most important reaction besides the H_2/CO sub-mechanism.

Starting from reaction 145, the important reactions show a diagonal-like distribution, i.e., the right half of Fig. 2. These reactions are fuel-related reactions and are listed in Table 2. Many of these reactions are hydrogen extraction reactions of the fuel, which is the first step in the whole combustion reaction process. This indicates that these reactions are important while having a large uncertainty and the potential to be optimized. This section can also provide another aspect of information: namely the diagonal-like

Table 2
Fuel-specific reactions with significant global sensitivity. The exemplar of each cluster exhibits distinct fuel-specific reactions.

Index	Reaction
145	$CH_3OH + H \leq\Rightarrow CH_2OH + H_2$
150	$CH_3OH + OH <=> CH_3O + H_2O$
190	$C_2H_2 + CH_3 <=> H + pC_3H_4$
211	$C_2H_3 + H <=> C_2H_2 + H_2$
212	$CH_3OH + OH <=> CH_3O + H_2O$
217	$C_2H_3 + O2 <=> CH_2CHO + O$
284	$C_2H_4 + OH <=> C_2H_3 + H_2O$
317	$C_3H_3 + H <=> pC_3H_4$
360	$H + aC_3H_5 (+M) <=> C_3H_6 (+M)$
369	$HO_2 + aC_3H_5 <=> C_2H_3 + CH_2O + OH$
398	$C_3H_6 + H <=> H_2 + aC_3H_5$
407	$C_3H_6 + OH <=> H_2O + aC_3H_5$
450	$C_3H_8 + H <=> H_2 + iC_3H_7$
520	$C_4H_6 + H <=> H_2 + iC_4H_5$
521	$C_4H_6 + H <=> C_2H_3 + C_2H_4$
529	$C_4H_6 + OH <=> H_2O + iC_4H_5$
597	$C_4H_81 + H <=> C_3H_6 + CH_3$
686	$C_4H_{10} + H <=> H_2 + sC_4H_9$

global sensitivity distribution justifies the clustering and exemplar selection. The original intention of using affinity propagation for clustering is due to the excessive amount of data that could not all be used for Bayesian optimization. While experimental conditions with similar dominant reactions are clustered into one class to reduce computational cost, the differences between the different classes should be reflected. This requires that the results of the affinity propagation algorithm should have two characteristics: (1) the exemplars in the class should be able to represent the characteristics of the vast majority of experimental conditions in the class, in this method i.e., the global sensitivity distribution; (2) the exemplars of different classes should reflect different characteristics to avoid redundancy. The reason for the second property is that when performing model optimization, the ideal situation is that the global sensitivity distributions of different experimental conditions can form a unit diagonal array. This means that only one reaction per condition is dominant and optimized, and this case has the highest optimization efficiency. However, a real reaction system cannot have only one dominant reaction, and the core mechanisms shared by different reaction systems will inevitably overlap. The sensitivity entropy proposed by Li et al. [27] and the differential entropy proposed by Sheen and Manion [28] and Valko et al. [29] follow the similar idea.

Results obtained through the current methodology demonstrate the second of the two characteristics described earlier by combining the analysis of the left and right halves. The left half comprises the shared core mechanism, which contains primarily the H_2/CO sub-mechanism and the C_1 species' mechanism, representing the common aspects of each category. On the other hand, the right half represents the fuel-specific reactions, corresponding to the distinctive part of each category. These reactions approximately form a diagonal arrangement such that various experimental conditions limit different reactions, ensuring effective optimization.

Owing to the high dimensionality of the data, with each data point characterized by a 784-dimensional global sensitivity under its specific conditions, visualizing the clustering results directly is impractical without further processing. To address this challenge, t-distributed stochastic neighbor embedding (t-SNE) is employed. t-SNE is a nonlinear dimensionality reduction technique that has gained prominence for its ability to embed high-dimensional data into a low-dimensional space, such as two or three dimensions, while preserving the underlying structure and relationships in the data. This makes t-SNE particularly suitable for visualizing complex datasets and identifying patterns or clusters within them. In this work, t-SNE is applied to reduce the dimensionality of the global sensitivity input data, transforming the original high-dimensional data into a more manageable 2-dimensional space (Fig. 3). This dimensionality reduction allows for a clearer and more concise visualization of the clustering results, making it easier to analyze and interpret the relationships between different experimental conditions and their respective clusters.

3.2. Uncertainty minimization

The preceding section confirms that exemplars from different clusters elicit distinct dominant reactions. However, it is uncertain whether the exemplars chosen through affinity propagation can effectively capture the overall uncertainty of their corresponding clusters, or if utilizing these exemplars in Bayesian optimization can effectively limit the model's uncertainty for the complete data set, encompassing the exemplars. Therefore, this section aims to optimize the uncertainty of the model parameters by employing the aforementioned 27 exemplars as experimental data, and subsequently assessing the model's performance on the data set as a whole via the posterior distribution.

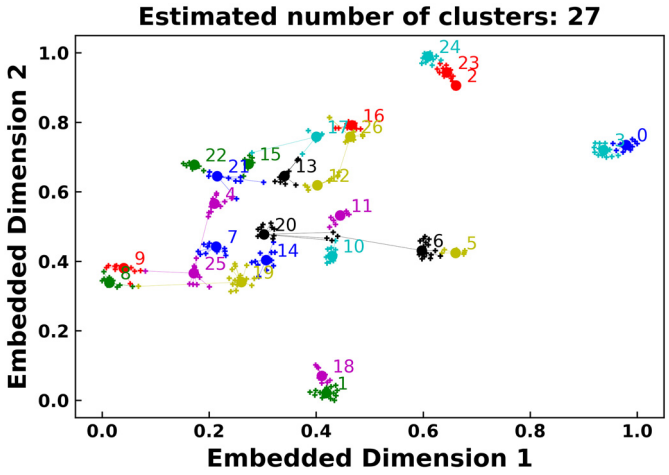


Fig. 3. Visualization of clustering results following dimensionality reduction using t-SNE. The exemplar of each cluster is represented by a solid dot, while the member of each cluster is denoted by a plus marker (+). The solid line illustrates the connection between the members and the exemplar within the same cluster. Each cluster is distinguished by a different color, and when necessary, colors may be reused due to a limited number of available options. The cluster index is indicated by the number in the upper right corner of each cluster. The associated experimental conditions are provided in Table 1.

The ANN-MCMC method is used to optimize the model parameters based on the selected exemplar experiment data. Since the present work proposes a clustering and representative selection method, it is more concerned with the clustering process and the uncertainty optimization capability of the overall data set when the exemplar is used as the optimization target. Consequently, the values of the experimental data are adopted from the model predictions to avoid potential problems of data consistency, which is not the focus of this method. Regarding the data uncertainty, the published experimental data on laminar flame speed have different definitions of uncertainty. Therefore, after comprehensive consideration of the error range that can be achieved by existing experimental techniques, the uncertainty is selected as 5 %, implying UF=1.05 for the experimental data. The model prediction uncertainties under the 27 selected exemplar experimental conditions are then constrained to 5 % following Bayesian optimization.

The union of activity parameters under 27 conditions contains 174 optimized parameters. The list of the most optimized reactions is shown in Table 3. The reactions that were highlighted in the global sensitivity analysis were effectively constrained. The prior and posterior parameter distributions of R118: 2 CH_3 <=> C_2H_5 + H and R597: C_4H_81 + H <=> C_3H_6 + CH_3 and

Table 3
The uncertainty constraint of dominant reactions in Bayesian optimization. Std is the posterior standard deviation of the parameters normalized by Eqs. (1) and (2). By definition, the standard deviation of the parameters before optimization is 0.5.

Index	Reaction	std
118	2 CH_3 <=> C_2H_5 + H	0.240
597	C_4H_81 + H <=> C_3H_6 + CH_3	0.281
108	CH_3 + HO_2 <=> CH_3O + OH	0.313
450	C_3H_8 + H <=> H_2 + iC_3H_7	0.324
686	C_4H_{10} + H <=> H_2 + sC_4H_9	0.324
0	H + O_2 <=> O + OH	0.339
150	CH_3OH + OH <=> CH_3O + H_2O	0.372
398	C_3H_6 + H <=> H_2 + aC_3H_5	0.391
165	HCCO + O_2 <=> 2 CO + OH	0.394
360	H + aC_3H_5 (+M) <=> C_3H_6 (+M)	0.397
317	C_3H_3 + H <=> pC_3H_4	0.401
521	C_4H_6 + H <=> C_2H_3 + C_2H_4	0.402
211	C_2H_3 + H <=> C_2H_2 + H_2	0.406

their joint probability distributions are given as examples in Fig. 4. The uncertainty in the pre-exponential factor of these reaction is constrained to 50 % of the original. It is worth noting that ordering of the constraint effect of uncertainty is not order-preserving with respect to the global sensitivities. Because the original model prediction uncertainty is different for different conditions, it leads to different constraint effects under the same data uncertainty (5 %).

Figure 5 shows the comparison of uncertainty factors before and after Bayesian optimization using data representing 27 experimental conditions. Uncertainties for all experimental conditions were constrained to within 40 %. The average value of the uncertainty factor decreased from 1.3 to 1.09. Using only no more than two conditions per fuel, results in a significant reduction in the model prediction uncertainty comparing to the original model for different pressures and equivalence ratio.

3.3. Discussion on model and data uncertainty

Experimental conditions with higher model prediction uncertainty and lower experimental data uncertainty are preferred. Upon reviewing Eq. (13), it can be observed that the net similarity comprises two distinct components:

$$\begin{aligned} NetSimilarity(c) &= \sum_{ik} c_{ik} s(i,k) \\ &= \sum_{i \neq k} c_{ik} s(i,k) + \sum_{k} c_{kk} s(k,k) \end{aligned} \tag{16}$$

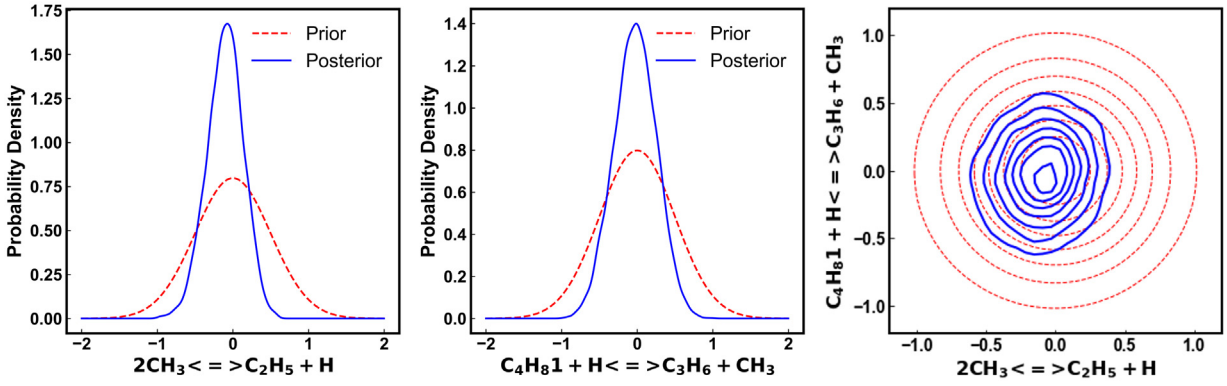


Fig. 4. Prior and posterior probability density distributions of the normalized pre-exponential factors of (a) R118: 2 CH_3 <=> C_2H_5 + H, (b) R597: C_4H_81 + H <=> C_3H_6 + CH_3, and (c) the two-dimensional joint distribution of the normalized pre-exponential factors of R118 and R597.

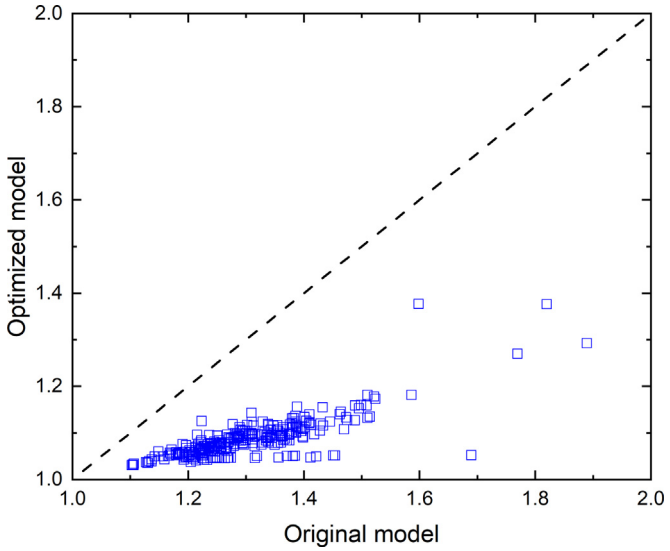


Fig. 5. Scatter plots of the uncertainty factors f_i under all experimental conditions in dataset. The dashed line is reference line where the original model and optimized model have equal uncertainty factors.

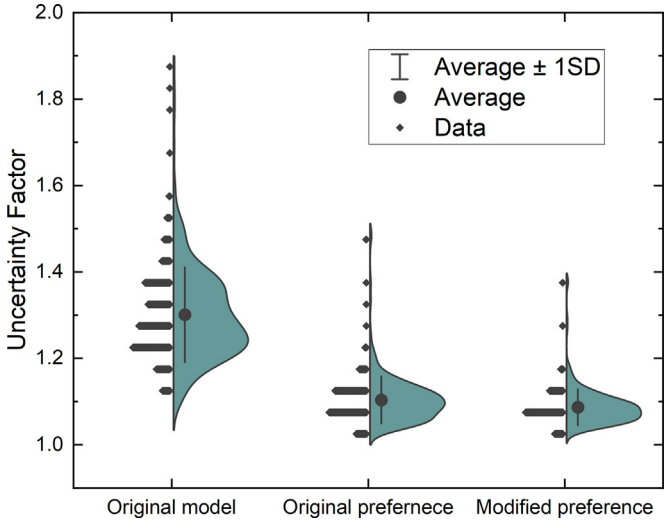


Fig. 6. Distribution of the uncertainty factors under all experimental conditions using the original and modified preference.

where the first item represents the sum of similarities between support points and exemplars, and the second item represents the sum of preferences of exemplars. The messaging process of affinity propagation is essentially maximizing the net similarity function. Consequently, when the preference of experimental conditions changes, experimental conditions with larger preference are more inclined to become exemplars in this cluster. With this theory, model and experimental uncertainties can be taken into consideration by increasing the preference of experimental conditions that meet the uncertainty requirements to obtain results that are more suitable for Bayesian optimization. As such, preference is multiplied by a factor of the ratio of experimental and model uncertainties. The modified preference is defined as:

$$s(k,k)' = s(k,k)\frac{UF_{exp}}{UF_{model}} \tag{17}$$

It is worth noting that this definition method is not unique. The custom preference provides more possibilities for the application of the algorithm in the combustion dataset. Potentially, more indicators of interest can be taken into account by changing the form of the formula and adding more items.

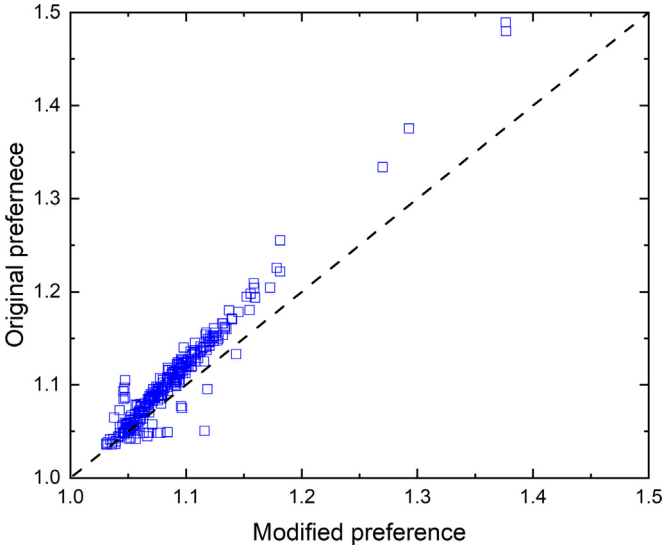


Fig. 7. Scatter plots of the uncertainty factors under all experimental conditions using different definitions of preference. The dashed lines are reference lines where the original model and optimized model have equal uncertainty factors.

Clustering and Bayesian optimization are conducted using the new preference definition. Fig. 6 shows the optimization result under this definition. The UF of all conditions is limited to less than 1.4, and the average value of UF drops significantly, from 1.3 to about 1.08, which is consistent with the set experimental data uncertainty. A comparison using the two types of preference is shown in both Figs. 6 and 7. Optimization using the new set of conditions results in an additional 2 % reduction in posterior uncertainty compared to the case where the prior model and experimental uncertainty are not considered. Given the strength of the previous constraint, the impact of the improved preference function was not significant because the model does not appear to have low priori uncertainty under the previous exemplars. However, as model optimization proceeds and larger datasets are used, the improved preference potentially has more significant advantages.

4. Conclusion

In this study, a method based on global sensitivity analysis and affinity propagation (GSAP) is proposed for experimental dataset clustering and exemplar condition selection. The datasets can be automatically divided into several clusters without specifying the number of clusters in advance. At the same time, an exemplar is selected in each cluster, which makes the cost of Bayesian optimization significantly lower if the optimization based on the exemplars can exert strong constraints on the entire dataset. Global sensitivity coefficients are used to characterize reactivity under different conditions.

Cluster analysis and exemplar selection are performed on a laminar flame speed dataset based on common C_0–C_4 fuels. 288 different experimental conditions are automatically clustered into 27 clusters, and 27 exemplars are given. By comparing the global sensitivity distributions of the exemplars of different clusters, it is clear that the clustering results are reasonable. The global sensitivities can be divided into two parts: the C_0–C_1 reactions shared by different fuels and the fuel-specific reactions. The fuel-specific reactions form an approximate diagonal array, which implies that there are different dominant reactions in different clusters and are

not redundant. Meanwhile, the results of t-SNE dimensionality reduction analysis corroborate the rationality of clustering.

Bayesian optimization is performed using 27 exemplary conditions. The optimization results show that the uncertainty factors of the exemplar conditions are all reduced to below 1.05. The average uncertainty of the entire data set is reduced from 30 % to 10 %, and the maximum uncertainty is constrained to below 50 %. The uncertainties of the highlighted reactions in the global sensitivity analysis are effectively bounded up to 50 %. This demonstrates that exemplars obtained by affinity propagation can effectively optimize the uncertainty of the overall dataset. In addition to the representativeness of the exemplars used for optimization, the uncertainty of the model predictions and experimental data under these conditions also needs to be considered. It is worth noting that, when there are multiple experimental data sources under given experimental conditions, based on the principle of uncertainty minimization, the data point with the smallest uncertainty will be selected for optimization. The new preference is defined as the original multiplied by the ratio of the experimental uncertainty to the model uncertainty. According to the principle of maximizing net similarity, conditions with larger model uncertainty and smaller experimental uncertainty tend to be exemplars. Although user-defined preference has not shown significant advantages in the current case, it could potentially be useful in more extreme cases. In addition to being used for Bayesian optimization, the exemplary dataset selected by the method can also be used as an optimal validation dataset in the model development process. Another potential application is experimental design. When the dataset consists of unexplored experimental conditions, the selected representatives correspond to the worthiest of conducting among the conditions, providing the most reaction kinetic information. This experimental design approach (GSAP) has been integrated into the OptEx platform [48,49].

Novelty and significance statement

This study presents an algorithm for clustering and representative experiment selection. The algorithm automatically clusters experimental conditions without the need to specify the number of clusters in advance. The algorithm is tested using a dataset on laminar flame speed. The results demonstrate a significant reduction in uncertainty across the entire dataset using only exemplars for optimization, which significantly reduce the computation cost. The potential of considering the model and experimental data was discussed.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Yiru Wang: Methodology, Investigation, Formal analysis, Writing – original draft. **Chenyue Tao:** Formal analysis. **Zijun Zhou:** Investigation. **Keli Lin:** Investigation. **Chung K. Law:** Writing – review & editing. **Bin Yang:** Conceptualization, Writing – review & editing, Supervision.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2023.113121.

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